

PETROLINA, Brazil

WMO: 829840

Lat: 9.362S

Lon: 40.569W

Elev: 1263

StdP: 14.04

Time Zone: -3.00 (W03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	66.3	67.8	46.6	49.1	89.7	48.6	53.0	89.0	20.6	78.3	18.9	78.8	7.0	140	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	20.5	98.9	70.7	97.2	70.4	96.7	70.3	76.8	85.9	76.0	85.6	75.3	85.3	8.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	138.1	78.7	73.6	131.3	78.5	73.1	128.9	78.6	41.2	86.4	40.4	84.5	39.7	85.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.5	16.8	15.7	DB	62.1	102.2	2.4	1.5	60.3	103.3	58.9	104.1	57.6	104.9	55.8	106.0
			WB	56.6	78.5	2.0	1.8	55.2	79.8	54.0	80.8	52.9	81.8	51.4	83.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	84.2	84.1	84.7	83.3	81.6	79.3	78.0	78.8	81.1	84.0	85.4	84.7	(6)
	DBStd	3.47	2.31	2.50	2.28	2.63	2.28	2.21	2.12	2.08	2.75	2.18	2.54	2.95	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	11833	1060	954	1077	998	978	880	867	893	934	1053	1063	1076	(10)
	CDD65	6358	595	534	612	548	513	430	402	428	484	588	613	611	(11)
	CDH74	71852	7268	6376	7471	6146	5322	3896	3402	4053	5271	7205	7973	7469	(12)
	CDH80	34962	3678	3178	3801	2901	2295	1526	1226	1692	2579	3812	4365	3909	(13)
(14) Wind	WSAvg	9.0	8.2	8.0	7.5	8.0	8.8	9.5	10.1	10.4	10.7	10.2	8.9	8.1	(14)
(15) Precipitation	PrecAvg	18.2	3.5	3.1	3.7	2.3	0.6	0.3	0.2	0.1	0.1	0.5	1.9	2.7	(15)
	PrecMax	39.8	16.9	14.6	14.6	11.7	4.1	2.0	1.2	0.8	0.8	5.0	7.7	18.3	(16)
	PrecMin	5.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	7.6	3.6	3.0	3.3	2.4	0.9	0.4	0.3	0.2	0.2	1.1	1.9	3.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.6	98.7	99.0	98.6	95.3	93.5	91.8	93.6	98.5	99.0	100.7	100.6	(19)
		MCWB	70.9	71.1	71.3	70.3	70.1	69.8	68.1	68.2	69.0	70.2	70.9	70.2	(20)
	2%	DB	96.8	96.8	97.1	96.5	93.4	91.6	89.8	91.6	95.3	97.0	98.7	98.4	(21)
		MCWB	70.2	71.1	71.0	70.6	70.7	69.1	68.1	67.5	68.5	69.7	71.0	70.6	(22)
	5%	DB	95.1	95.1	95.4	93.6	91.6	89.7	88.0	89.8	93.3	95.2	96.9	96.6	(23)
		MCWB	70.4	71.4	71.8	71.9	70.8	69.3	67.8	67.6	68.1	69.3	70.6	70.4	(24)
	10%	DB	93.4	93.2	93.5	91.6	89.8	87.8	86.3	88.0	91.3	93.4	95.1	94.7	(25)
		MCWB	70.5	71.6	72.1	72.0	71.2	69.1	67.8	67.0	67.8	69.1	70.5	70.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.6	77.1	77.3	77.4	76.4	73.8	73.4	72.0	73.0	74.7	76.3	77.3	(27)
		MCDB	83.2	87.6	87.3	86.4	85.6	85.8	82.2	84.3	89.1	86.9	85.7	85.4	(28)
	2%	WB	76.3	76.0	76.3	76.0	74.7	72.8	71.9	70.9	71.4	73.7	74.8	75.6	(29)
		MCDB	82.8	86.4	87.0	85.4	84.5	84.1	81.4	83.4	86.0	87.6	88.1	85.3	(30)
	5%	WB	75.0	75.1	75.4	75.1	73.9	71.9	70.9	69.9	70.5	72.6	73.9	74.5	(31)
		MCDB	82.6	85.1	86.6	84.8	84.1	82.4	81.1	82.9	85.6	86.9	88.2	85.4	(32)
	10%	WB	73.6	74.3	74.6	74.3	73.2	71.1	70.0	68.9	69.8	71.5	72.9	73.6	(33)
		MCDB	85.1	84.9	86.0	84.4	83.7	81.5	80.8	81.8	85.2	85.8	87.9	85.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	18.7	19.2	18.1	17.7	18.2	18.5	20.5	21.4	21.0	20.5	19.2	(35)
		MCDBR	20.9	20.7	21.2	20.5	20.1	20.8	20.4	22.2	23.6	22.4	22.1	21.9	(36)
	5% WB	MCWBR	5.9	5.5	5.9	5.8	5.5	5.9	5.3	6.0	6.7	6.1	6.9	6.9	(37)
		MCWBR	17.1	17.5	17.9	16.7	16.8	17.7	17.4	20.1	21.0	19.8	19.5	17.7	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.389	0.382	0.383	0.372	0.354	0.347	0.337	0.352	0.378	0.407	0.407	0.404	(40)
		taud	2.472	2.506	2.506	2.528	2.591	2.585	2.607	2.508	2.416	2.371	2.393	2.413	(41)
	Ebn at Noon	Ebn at Noon	302	302	297	291	288	285	290	293	294	291	294	297	(42)
		Edh at Noon	37	36	36	33	30	29	29	33	38	41	40	39	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1986	1901	1867	1739	1572	1516	1598	1864	2084	2127	2067	1980	(44)
	RadStd	RadStd	169	124	104	113	113	87	77	71	91	110	131	123	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page